BG-Flood: A GPU adaptive, open-source, general inundation hazard model

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Abstract
Numerical modelling is a key tool for forecasting and assessing coastal, fluvial and intense rainfall inundation hazard. Inundation hazard, however, is usually driven by several factors such as storm surge, waves (setup/runup), rainfall and river flooding occurring at the same time. Simulating all these drivers at a relevant scale can be challenging and computationally intensive. To alleviate some of these challenges, we developed, BG-Flood (**B**lock-adaptive, **G**PU capable, **Flood** model), a numerical model for simulating shallow water hydrodynamics on Graphics Processing Units (GPU) using adaptive quad-tree type mesh. The model is designed with the goal of simulating inundation from any flood driver (rainfall, river, storm surge or tsunami) where the quad-tree mesh is automatically generated and refined by the model using cartesian grids as forcing (i.e. for DEM, wind, rainfall). The model is designed to use various mesh refinement strategies allowing the user to guarantee high resolution in the area of interest without the need to manually manipulate the model mesh but also facilitate automation of model pre- and post-processing.

Although the model is still under development, it offers an open-source solution to simulate inundation hazard seamlessly across multiple geographical scales.

Keywords: Modelling, open-source, inundation, GPU, AMR.

1. Introduction

Physics-based numerical modelling is a key tool for forecasting and assessing inundation hazards whether from coastal drivers, fluvial floods or intense rainfall ([1], [18]). These physics-based models often solve the non-linear Shallow Water Equations (Saint-Venant Equations) or dispersion-capturing models such as the higher order Boussinesq/Green-Naghdi Equations or non-hydrostatic multi-layer versions of the Shallow Water Equations.

While computers and model implementations have become more efficient, the widespread availability of high resolution (i.e. finer than 10m) Digital Elevation Models (DEM) drive demand for larger domains and finer resolution models [18]. In addition, there is a demand for running an increasing number of simulations for (1) covering a wide range of forcing to inform planning decision (e.g. adaptive planning), (2) testing the sensitivity of the model setup, (3) ensemble forecasting and (4) probabilistic assessments. There is therefore a strong pressure for models to become more computationally efficient and make the most of available computational resource.

Modern inundation models are also expected to be capable of simulating several factors coinciding such as storm surge and waves (setup/runup), rainfall and river flooding occurring at the same time. This is particularly true for numerical models designed for coastal application.

Simulating all these drivers at a relevant scale can be challenging and computationally intensive and even more so when combining these drivers in the same model. Several strategies can be applied to reduce the computation burden:

(1) Optimise the number of grid/mesh nodes using flexible/unstructured mesh (typically made with triangle) or quadtree type mesh [13]. The solvers for these complex mesh layouts increase the computational cost per mesh node (when compared to simple Cartesian model) but allow to significantly reduce the overall number of nodes to accurately represent the flow dynamics. However, designing unstructured mesh can require a significant user input.
(2) Offload all or part of the computation to Graphics Processing Units (GPU), which involve a redesign of the model memory layout to optimise the occupancy[1] of the GPU. These initiatives are becoming common in research code (e.g. LISFLOOD-FP [15]) and have since been integrated to commercial models such as TUFLOW [12] and Mike 21 [8]. However, it is uncommon in community or open-source models.
(3) Simplify the solution by using implicit solver and/or neglect terms in the equation (i.e. reduced physics). While these simplifications lead to great reductions in computational cost they can also lead to inaccuracies or even

[1] Ratio of active number of warps (group of threads) to the maximum available

aberrant results when used outside the validity of these assumptions [9] and may be undesirable for coastal application.

Unfortunately, there only exist few examples of open-source tools that include optimised grid layout mesh with computation performed on the GPU.

This article describes BG-Flood (Block-adaptive, GPU capable, Flood model), a numerical model for simulating shallow water hydrodynamics on GPU using adaptive quad-tree type mesh (Figure 1). The model is designed with the goal of simulating inundation from any flood driver (rainfall, river, storm surge or tsunami) on an optimised mesh that is automatically generated and refined by the model using cartesian grids as forcing (i.e. for DEM, roughness, wind, rainfall).

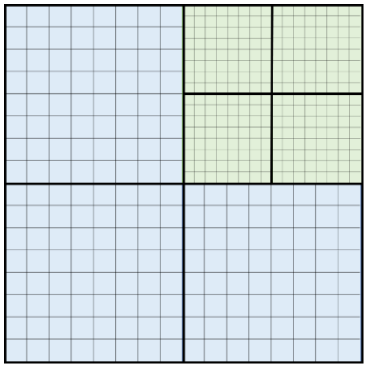

Figure 1 Example of BUQ grid with 8x8 cells blocks and 2 resolution levels

The model core principals are briefly described in Section 2, Section 3 shows the model validation to three benchmark tests and section 4 gives two inundation examples, and shows the use of a simple and yet efficient mesh refinement strategy. This allows the user to guaranty high resolution in areas of interest without the need to manually manipulate the model mesh but also facilitate automation of model pre- and post-processing.

2. BG-Flood Model description

BG-Flood simulates inundation by solving the shallow water equations (also referred to Saint-Venant Equations). BG-Flood borrows the open-source finite volume method that is part of the Basilisk framework [14] that solves the full Shallow Water Equations. However, the memory model is modified to follow the Block Uniform Quadtree (BUQ) implementation of [19].

The model interface is through a simple text file that BG-Flood reads upon start up. While not as engaging as a graphical user interface the text document is a good compromise for a human readable file that can also easily be automatically generated from scripts.

2.1 Cartesian Input forcing

Input for spatially varying forcing (DEM, wind, rain) is done through one or more Cartesian grids. In case of DEM, the first grid given to the model is used to initialise the computational mesh, other grids are used to describe the DEM at finer resolution and to calculate elevation at refined mesh nodes.

2.2 Computational mesh: BUQ mesh

BUQ meshes are a compromise between Cartesian grids (easy to optimise for GPU) and quadtree meshes (difficult to optimise for GPU). In a BUQ mesh, there is a quadtree structure that is made up of smaller Cartesian blocks. Each block contains the same number of nodes (e.g. 8x8) but each block is not necessarily at the same resolution. In an adaptive BUQ mesh, a block can be divided into four child blocks each with half the resolution of the parent block (see

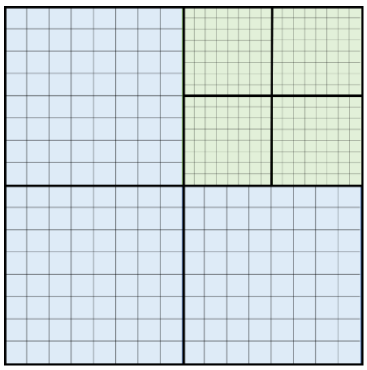

Figure 1). Each resolution corresponds to a level of refinement where:

$$dx = dx_0 \times 2^{-level} \quad (1)$$

where dx is the effective resolution for the block, dx_o is an arbitrary reference resolution and *level* is the level of refinement relative to the reference level (level 0). This formulation allows for using negative (or positive) levels when the level is coarser (or finer respectively) than the reference resolution.
BG-Flood automatically generates the BUQ mesh used in the computation by the following process:

(1) Generate an initial cartesian grid with a size proportional to the number of elements per block and an initial resolution (Equation 1)
(2) Generate BUQ mesh from cartesian grid at an arbitrary initial level
(3) Calculate initial (water level) condition
(4) Iterate a finer/coarser adaptive strategy (Figure 1) until convergence
(5) Recalculate initial condition

2.3 Roughness formulation

Three friction formulations are available: Quadratic, Manning and Smart [16] formulation. The Manning formulation is available for compatibility with other models, but the Smart formulation is recommended for shallow flow.

2.4 Limitation

The model formulation is subject to the typical dispersive property of the Shallow Water Equations.

Reconstruction of the second order term can lead to spurious velocities in shallow flow over steep topography. To limit this issue the model uses the Buttinger et al. [5] formulation.

3. Validation

The core engine of BG-Flood is borrowed from the Basilisk framework [14] and has been previously verified using standard benchmarks. Below benchmarks are provided for confirmation that BG-Flood produces accurate results for typical inundation applications.

3.1 Tsunami – Monai Valley benchmark

The "Monai Valley benchmark" is well known and a standard tsunami benchmark recommended as part of model validation and verification. The benchmark is a laboratory model closely resembling the actual bathymetry and topography of the Monai Valley in Okushiri Island, Japan, where the 1993 tsunami resulted in extreme runup. The laboratory experiment reproduces the extreme runup at 1/400 scale. A description of the model is given in [17].

BG-Flood can reproduce the benchmark wave accurately at the 3 nearshore gauges (Figure 2) although, as with other shallow water equation models, it does not reproduce some of the high frequency features of the wave.

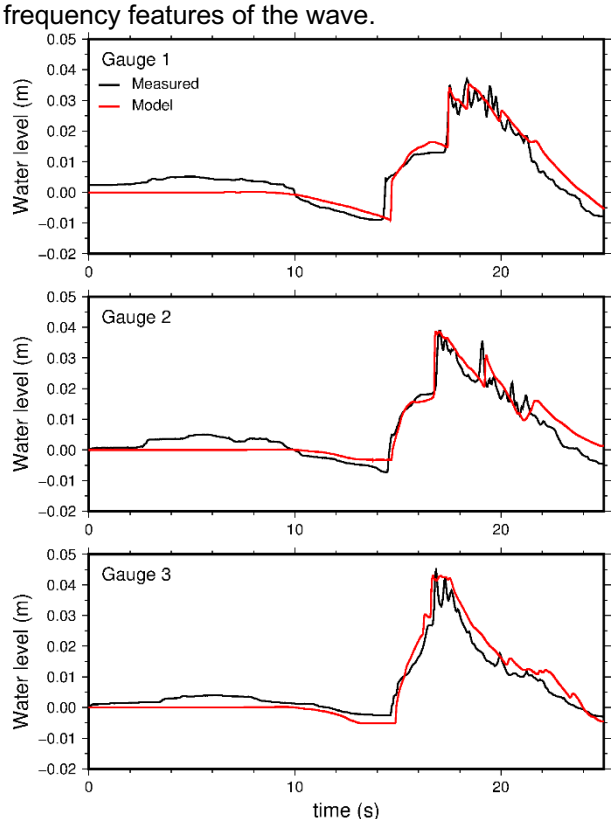

Figure 2 Comparison of BG-Flood against the Monai Valley benchmark. The black line shows the measured values and the red line the simulated results.

3.2 Tsunami Currents – Tauranga Tohoku

Observations of the 2011 Tohoku tsunami reaching Tauranga harbour provide an excellent validation benchmark for simulating tsunami induced currents. The benchmark is fully described in [2] and [3].

The setup for this benchmark is similar to the Benchmark problem 3 from [7]. The bathymetry has been rotated so that the shore is parallel to the top boundary which is forced with the Beacon A tsunami record. The incident waves correspond to the complete signal and include both the tsunami and the tide signal.

BG-Flood adequately reproduces the water levels and velocity magnitudes from the ADCP location (Figure 3). While this test validates the use of Shallow Water Equations for simulating tsunami propagation in harbours and estuaries it is also a verification for the non-reflecting model boundaries. Given that the forcing boundary is relatively close to the shore and observation point, a reflected wave at the boundary or a faulty implementation would result in a quick contamination of the domain by reflections.

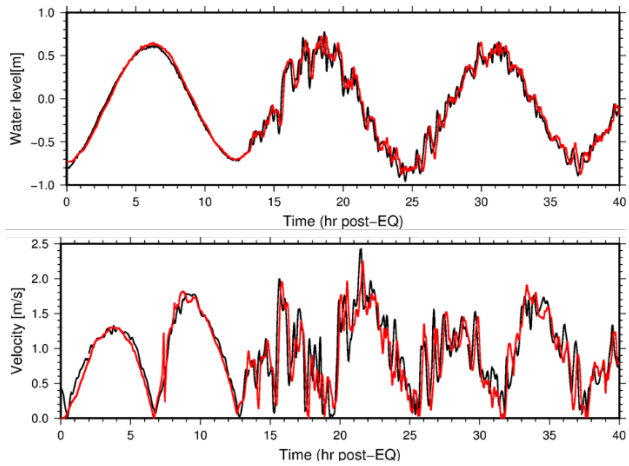

Figure 3 Tauranga Harbour entrance ADCP during the 2011 Tohoku tsunami. The black line shows the measured observations and the red line the simulated results for water level (top panel) and velocity magnitude (lower panel).

3.3 Rainfall Over a Two-dimensional Experimental Catchment

The laboratory experiment of [6] is used to assess the model performance for shallow water flow generated by uniform rainfall on an arbitrary V shape basin with walls on the slopes. The model results for case C1 are presented in Figure 4.

The experiment presents several challenges with experimental variables that are difficult to constrain and cannot easily be included in the model (leaky nozzles, surface tension in very shallow flow and how the experiment achieved rainfall uniformity). Therefore, a perfect comparison between the model and experimental results is not expected here. In

fact, the experiment results show oscillations in the flux outlet due to natural oscillations occurring in the experimental basin. These oscillations offer a great challenge for model to capture.

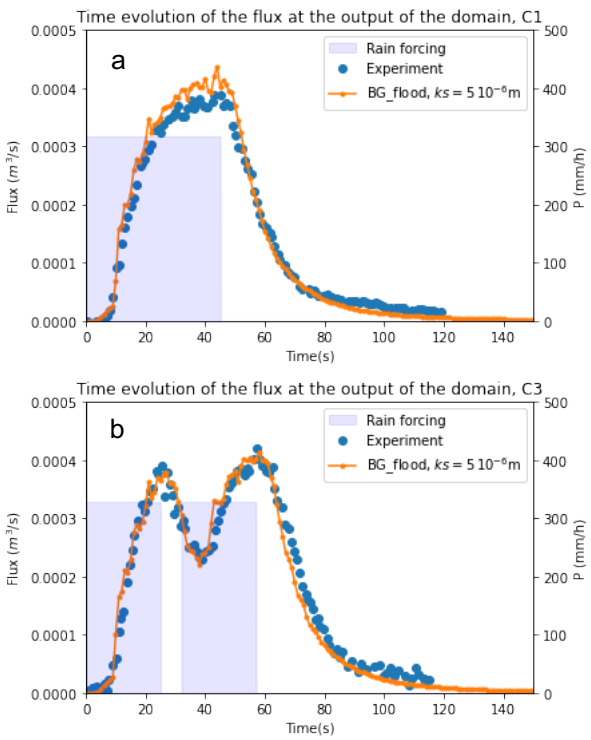

Figure 4 Cea et al. [6] experiment for 2D rainfall runoff tests. Experiment C1 (a) and experiment C3 (b).

BG-Flood accurately captures the outlet hydrograph of the experiment. The model also captures the amplitude and period of the oscillation in the flux suggesting that it also captures the oscillation of the basin upstream. This feature is missed in some model (e.g. [6] and [21])

4. Examples of realistic configurations

BG-Flood adaptive mesh capability is demonstrated using two examples of inundation: modelling the effects of the 2009 South Pacific Tsunami in South-East Upolu Samoa and a river flood in the Waikanae Catchment in Kāpiti Coast of New Zealand coinciding with a high storm-tide level. In each model, the mesh refinement is guided by results from an initial coarse scale cartesian model.

Table 1 Table of the adopted fault parameters for the three potential rupture scenarios of the 2009 South Pacific Tsunami

Rupture segments	Length / Width (Km)	Strike/Dip/Rake Angle (Deg)	Slip (m)	Centroid of Rupture Depth Long. Lat.	Rupture Timing Start / Rise Time (s)
1	130 / 50	324 / 65 / -85	10.2	18.0 / 172.0 / -15.5	3.0 / 60.0
2	50 / 50	175 / 29 / 90	4.7	18.0 / 172.4 / -16.0	49.0 / 40.0
3	50 / 50	180 / 29 / 90	4.7	18.0 / 172.4 / -16.0	90.0 / 40.0

4.1 2009 South Pacific Tsunami in Samoa

Regional tsunami inundation can be challenging to simulate because it requires a high resolution for the inundation area of interest but also a sufficiently large computational domain to capture the offshore tsunami wave evolution. In this example, we simulate the generation and propagation of the 2009 South Pacific Tsunami (SPT) and focus on inundation on the south-east coast of Upolu in Samoa where runup levels were the greatest.

4.1.1 Datasets

The 2009 SPT was initiated by a complex earthquake that involved rupture on multiple faults in the subduction zone and on the outer rise. In this simulation, we model the tsunami generation using three surface deformation patches corresponding to different regions of fault rupture that occur sequentially. This scenario was initially modelled in [4] based on results from [10] (**Error! Reference source not found.**). The model is driven by 3 DEMs: a 250 m downscaled GEBCO bathymetry of the South Pacific (), a 50 m DEM of the Samoa region created by merging the GEBCO bathymetry, multi-beam survey and bathymetry/topography LiDAR; and a 10 m DEM of the South-East Upolu created by combining LiDAR topography and multibeam bathymetry.

For each rupture segment a deformation field is created using the 1985 Okada deformation algorithm [11][2].

4.1.2 Iterative refinement

The complex behaviour of tsunami waves including refraction, reflection, and resonance, can often lead to high runup in unexpected places. It is therefore difficult to estimate areas where the model will require higher resolution or where coarser grid cells may be suitable without first running the model. Additionally, high resolution may be needed in different locations at different times, as when a tsunami wave crosses the Pacific Ocean. Ideally, in this case, the resolution of the grid follows the wave. In this example we first run the model at a medium resolution of 500 m and use the offshore wave height from this simulation to guide our refining

[2] using the Julia Tsunami package see: https://github.com/CyprienBosserelle/Tsunami.jl

criteria. Our area of interest (South-East Upolu) is also used to constrain the highest level of refinement.

Figure 5 shows the refined mesh at the regional scale and **Error! Reference source not found.** shows the mesh at the South-East Upolu scale.

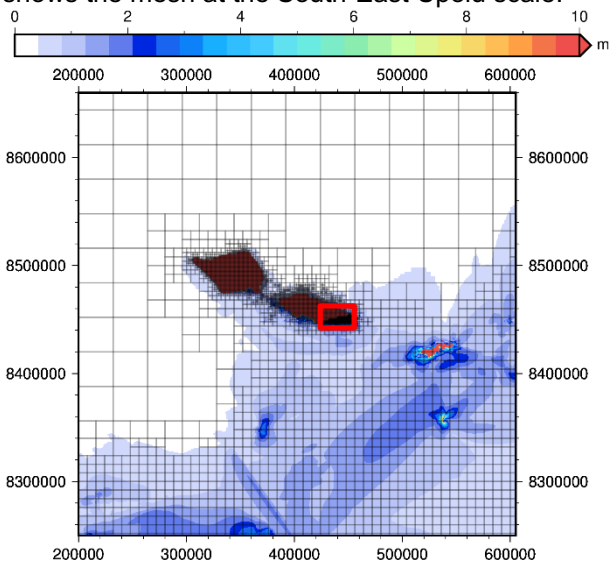

Figure 5 Modelling of the 2009 SPT at the regional scale. The BUQ mesh shows different levels of resolution in different areas of the domain. Maximum wave height for the modelling is shown in shading over the ocean. The red square corresponds to the domain in Figure 6.

The adaptive model runup (**Error! Reference source not found.**) gives similar results to the traditional nested model method [4]. The overall refined mesh simulation was also significantly faster at 3.5 h compared to the 6 h required for the nested run from [4]. The adaptive mesh also followed a strict rule that ensures there is only one level of refinement between neighbouring blocks. Meanwhile the nested simulation [4] jumps from 250 m resolution to 50 m and then to 10m. This could explain the lower runups from [4].

4.2 Waikanae River Flood

Many of the NZ rivers traverse the low-lying coastal plain before reaching the ocean. If the flood peak is of short duration and coincidse with low tide, a significant flood can have little consequence, whereas if the same flood coincides with a high tide or a high storm tide, the impact can be much greater. In this example we simulate an arbitrary 24 h flood event on the Waikanae catchment with the peak of the flood reaching the outlet at low tide. In order to adequately represent the elevation of stop-banks along the river in the model, the mesh requires a high-resolution (~5 m).

4.2.1 Iterative model

For this example, a script was used to first simulate the model on a simple coarse cartesian mesh (i.e. a BUQ mesh with a single level). Using simple arithmetic operations, the area of maximum flood extend was extracted and used to produce a target refinement levels grid to guide BG-Flood in the creation of a finer grid.

The DEM input to the model in all simulations is a 4.5 m DEM constructed from the regional LiDAR. The roughness map is a roughness length calculated from the raw LiDAR point cloud and gridded on a 4.5 m grid with the same extent as the DEM.

The initial coarse 50 m resolution mesh is automatically constructed by BG-Flood using the 4.5 m DEM and inputs are automatically recalculated for that mesh. Results of the coarse simulation are quickly obtained (8 min simulation time on a mid-range Quadro P620 GPU). Maximum flow depth at the end of the simulation is used as a fag for a level 4 refinement (flood extend as well as river bed location) as well as the shallow depth (less than 0.5 m) at the start of the simulation (nearshore area). This step is done using a single command to the grdmath module of the Generic Mapping Tools (GMT) [20]. Using the target level BG-Flood constructs the BUQ mesh with level 4 blocks (5 m resolution) on the expected flood area and shoreline gradually coarsening to level 0 (80 m resolution) away from the flood area.

Final simulated flow depth and block layout are presented in Figure 7 for the lower part of the river and for a specific section of the river in Figure 8. The outlines of blocks have also been drawn to show the

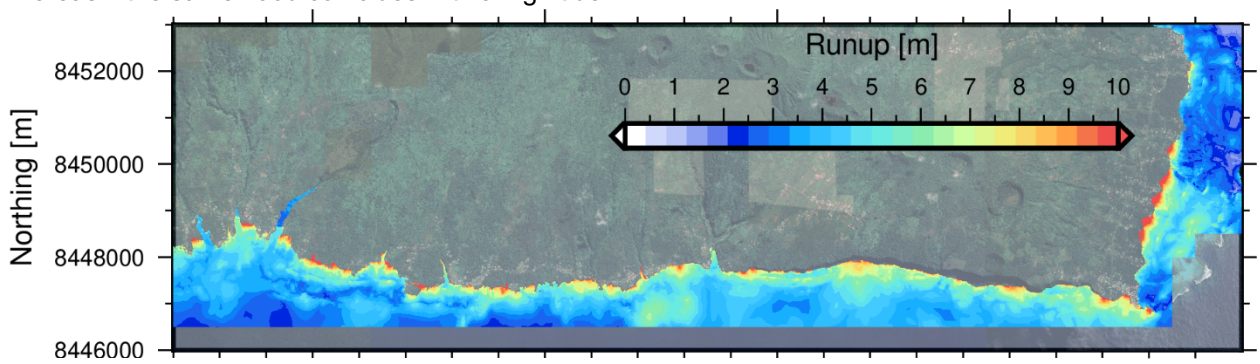

Figure 6 Close up runup modelling for the South East coast of Upolu. Only highest-level outputs (level 4) are shown.

result of the automatic mesh refinement, each block containing 16x16 cells in this example.

In the model setup the strategy for refining the mesh used an initial iteration to select the interesting part of the model where the final model requires its highest refinement. The target level grid created from the coarse model results could be created using more complex strategies depending on the user requirement. It is easy to imagine a mesh with an even finer level of refinement where the flow velocity or water depth exceed a certain threshold.

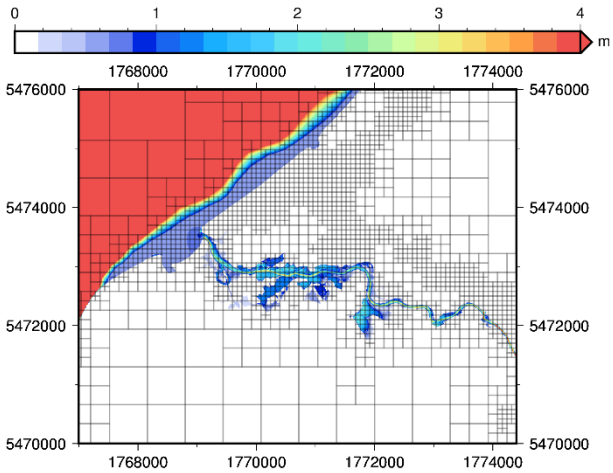

Figure 7 Model block outline (gray squares) and maximum flow depth (shading) for the lower part of the Waikanae River

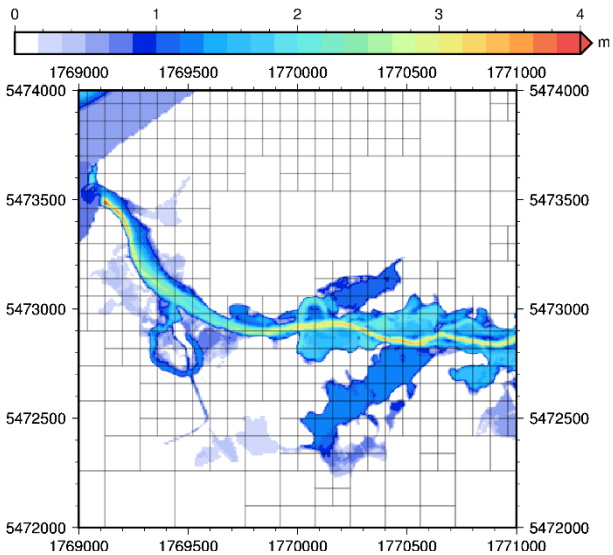

Figure 8 Model block outline (gray squares) and maximum flow depth (shading) for the lowest part of the Waikanae River.

5. Summary

BG-Flood is a numerical model for simulating shallow water hydrodynamics on the GPU using an BUQ adaptive mesh. The model is open source (C++/CUDA C++) and free and can be used for a range of coastal, river and rainfall flooding. The model is still in early stage of development but is already potentially useful for practitioners for simulation of flood near the coast.

While this article presents a selected number of benchmark verifications of the model, a full set of verification is being prepared for the model repository as well as a collection of internal tests for continuous testing. The continuous testing can be used to give confidence for novice developers that their customisation of the code doesn't adversely affect the model accuracy/performance.

The code and Windows executable are publicly available at: https://github.com/CyprienBosserelle/BG_Flood

Code documentation and wiki are available at: https://bg-flood.docsforge.com/

Users are also encouraged to submit comments and request to the GitHub page.

6. Future developments

The model development is currently focused on creating internal test and eradication of unwanted behaviours. In the next months the model development will also focus on:

- Adaptive mesh during simulation
- Optimisation of GPU kernels
- 1D Flow structures (bridges and culverts)

Long-term development goal may also include:

- Multi-layer non-hydrostatic equations for tsunami and wind waves
- Group resolving wave forcing

7. Acknowledgement

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